



Aditya Raj Pandey
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Education	Specialization	Institute	Year
B.Tech	Engineering Physics	IIT Delhi	2025

SCHOLASTIC ACHIEVEMENTS

- JEE Advanced 2021: AIR 4326; JEE Mains 2021: AIR 6451.

PROFESSIONAL EXPERIENCE

- **Software Engineer - AI/ML | CheQ Digital Pvt. Ltd, Financial Services Platform | Bengaluru** *Sept. '25 – Present*
 - Developed AI solutions for finance, covering credit, merchant intelligence, and compliant decision support.
 - **Data Annotation Project:- In-House Merchant extraction from transactions model development & deployment**
 - * Led POC in-house **Merchant NER**, categorizing transactions, eliminating vendor reliance in regulated finance.
 - * Benchmarked and fine-tuned **SpaCy, GLiNER, FinBERT** with model evaluation and compliance validation.
 - * Industrialized **FinBERT** with **data scaling, BIO→BILO migration**, optimization, and production deployment.
 - * Achieved a **97% F1-score** with the scaled **FinBERT** model and deployed it to the production assembly line.
 - **Sentiment Classifier Project:- Prompt-Based User Query Sentiment & Issue Classification**
 - * Led problem scoping with Product, defining taxonomy and ownership through cross-functional collaboration.
 - * Designed sentiment and issue classifier using **advanced prompt engineering techniques** & iterative refinement.
 - * Delivered user-sentiment intelligence via alerts, logging, enabling product improvement and decisions.
 - **Chatbot Web Retrieval Project:- Domain-Specific Crawling**
 - * Led POC evaluating **Firecrawl, Exa, Vertex AI, Crawl4AI** for chatbot web search integration.
 - * Analyzed 3 months of chatbot web queries; created a **50-query golden dataset** covering all types.
 - * Built curated pipeline performing weekly crawling, **LLM summarization & vector database embedding** storage.
 - Collaborated with **Product** and **Testing** teams, iterating **ML systems**, ensuring **SDLC process alignment**.
 - Managed **stakeholders** across functions, driving **model development, deployment, and operational improvements**.

Projects

- **LLM-Powered SQL Chatbot: Generative AI, Prompt Engineering, Full-Stack Development | Self Project**
 - Built **LLM-based chatbot** using **FastAPI** to convert natural language into executable **SQL database queries**.
 - Integrated **Groq-hosted Llama 3 model** with **PostgreSQL**, enabling **AI-driven real-time query translation**.
 - Developed secure **ReactJS frontend** with formatted response display, logging, and **error handling features**.
- **Cyclone Preheater Anomaly Detection: Unsupervised ML | Anomaly Detection | DBSCAN | Self Project**
 - Analyzed high-frequency sensor data (370k+ entries, 3 years) using **DBSCAN** to detect time-series anomalies.
 - Applied **Min-Max Scaling**, handled non-numeric values, tuned clustering via **k-distance** and **silhouette scores**.
 - Detected anomalies in pressure and temperature signals; visualized system-wide failures and localized deviations.
- **Movie Box Office Data Scraper: Web Scraping | ETL | Data Engineering | Self Project**
 - Developed a web scraper with **BeautifulSoup** to extract box office data for **3000+ movies** from thenumbers.com.
 - Implemented parallel processing via **ThreadPoolExecutor** for faster scraping of large datasets.
 - Built an **ETL pipeline** to clean, transform, and load movie data into CSV files, integrating with **TMDB metadata**.

Technical Skills

Programming & Data Handling: Python, Java, JavaScript, SQL, Data Structures & Algorithms, Python Data Libraries (NumPy, Pandas, Matplotlib, Seaborn), Advanced Excel, BeautifulSoup, ThreadPoolExecutor
Machine Learning & GenAI: PyTorch, Scikit-learn, TensorFlow, GPT-4, LangChain Agents, FAISS Vector Store, RAG, Prompt Engineering, Sentence Transformers, TF-IDF Vectorization, Prompt Chaining, LLM Evaluation, Naive Bayes, Logistic Regression, KNN, Decision Trees, CNNs, Cosine Similarity, Sentiment Analysis, Feature Engineering, Hyperparameter Tuning
Tools & Platforms: LangChain, Streamlit, OpenAI API, Hugging Face Transformers, Unstructured.io API, Power BI, Microsoft Office
Soft Skills: Cross-functional collaboration, stakeholder management, agile teamwork, effective communication, problem-solving, Risk assessment, Model governance

Positions of Responsibility

Logistics Coordinator (RendezvousX, Sept 2023 – Oct 2023):Led a dedicated **10-member team** managing comprehensive end-to-end logistics for **3 major cultural events**, ensuring seamless and flawless execution.
Events Coordinator (PhySoc, Aug 2023 – Apr 2024):Organized events including **Intern Insights, faculty-student meetups**,etc.Led **cross-functional marketing & publicity team** to enhance engagement & collaboration.
Events Executive (Tryst'23, Dec 2022 – Mar 2023):Led initiatives securing sponsorships from **Quickheal** and **Neovel**, orchestrated 6 successful events with **CXO-level collaboration**, and achieved **700+ registrations**